

19 Nested vectored interrupt controller (NVIC)

19.1 NVIC features

The NVIC includes the following features:

- up to 150 maskable interrupt channels for STM32H7xxx (not including the 16 interrupt lines of Cortex[®]-M7 with FPU)
- 16 programmable priority levels (4 bits of interrupt priority are used)
- low-latency exception and interrupt handling
- power management control
- implementation of system control registers

The NVIC and the processor core interface are closely coupled, which enables low latency interrupt processing and efficient processing of late arriving interrupts.

All interrupts, including the core exceptions, are managed by the NVIC.

For more information on exceptions and NVIC programming, refer to PM0253 programming manual for Cortex[®]-M7.

19.1.1 SysTick calibration value register

The SysTick calibration value (SYST_CALIB) is fixed to 0x3E8. It provides a reference timebase of 1 ms based when the SysTick clock frequency is 1 MHz. To match the 1 ms timebase whatever the application frequency, the SysTick reload value must be programmed as follows in the SYST_RVR register:

- The SysTick clock source is the 100 MHz CPU clock (HCLK):

$$\text{reload value} = (F_{\text{HCLK}} \times \text{SYST_CALIB}) - 1$$

- or the SysTick clock source is an external clock:

$$\text{reload value} = ((F_{\text{HCLK}}/8) \times \text{SYST_CALIB}) - 1$$

where F_{HCLK} refers to the AHB frequency expressed in MHz.

For example, to achieve a timebase of 1 ms when the SysTick clock source is the 100 MHz HCLK:

$$\text{reload value} = (100 \times \text{SYST_CALIB}) - 1 = 0x1869F$$

19.1.2 Interrupt and exception vectors

The exception vectors connected to the NVIC are the following: reset, NMI, HardFault, MemManage, Bus Fault, UsageFault, SVCcall, DebugMonitor, PendSV, SysTick.

Table 143. NVIC⁽¹⁾

Signal	Priority	NVIC position	Acronym	Description	Address offset
-	-	-	-	Reserved	0x0000 0000
-	-3	-	Reset	Reset	0x0000 0004
-	-2	-	NMI	Non maskable interrupt. The RCC Clock Security System (CSS) is linked to the NMI vector.	0x0000 0008
-	-1	-	HardFault	All classes of fault	0x0000 000C
-	0	-	MemManage	Memory management	0x0000 0010
-	1	-	BusFault	Prefetch fault, memory access fault	0x0000 0014
-	2	-	UsageFault	Undefined instruction or illegal state	0x0000 0018
-	-	-	-	Reserved	0x0000 001C-0x0000 002B
-	3	-	SVCcall	System service call via SWI instruction	0x0000 002C
-	4	-	DebugMonitor	Debug monitor	0x0000 0030
-	-	-	-	Reserved	0x0000 0034
-	5	-	PendSV	Pendable request for system service	0x0000 0038
-	6	-	SysTick	System tick timer	0x0000 003C
wwdg1_it	7	0	WWDG1	Window Watchdog interrupt	0x0000 0040
exti_pwr_pvd_wkup	8	1	PVD_PVM	PVD through EXTI line detection interrupt	0x0000 0044
exti_tamp_rtc_wkup	9	2	RTC_TAMP_STAMP_CSS_LSE	RTC tamper, timestamp	0x0000 0048
lsecss_rcc_it				CSS LSE	
exti_wkup_rtc_wkup	10	3	RTC_WKUP	RTC Wakeup interrupt through the EXTI line	0x0000 004C
flash_it	11	4	FLASH	Flash memory global interrupt	0x0000 0050
rcc_it	12	5	RCC	RCC global interrupt	0x0000 0054
exti_exti0_wkup	13	6	EXTI0	EXTI Line 0 interrupt	0x0000 0058
exti_exti1_wkup	14	7	EXTI1	EXTI Line 1 interrupt	0x0000 005C
exti_exti2_wkup	15	8	EXTI2	EXTI Line 2 interrupt	0x0000 0060

Table 143. NVIC⁽¹⁾ (continued)

Signal	Priority	NVIC position	Acronym	Description	Address offset
exti_exti3_wkup	16	9	EXTI3	EXTI Line 3 interrupt	0x0000 0064
exti_exti4_wkup	17	10	EXTI4	EXTI Line 4 interrupt	0x0000 0068
dma1_it0	18	11	DMA_STR0	DMA1 Stream0 global interrupt	0x0000 006C
dma1_it1	19	12	DMA_STR1	DMA1 Stream1 global interrupt	0x0000 0070
dma1_it2	20	13	DMA_STR2	DMA1 Stream2 global interrupt	0x0000 0074
dma1_it3	21	14	DMA_STR3	DMA1 Stream3 global interrupt	0x0000 0078
dma1_it4	22	15	DMA_STR4	DMA1 Stream4 global interrupt	0x0000 007C
dma1_it5	23	16	DMA_STR5	DMA1 Stream5 global interrupt	0x0000 0080
dma1_it6	24	17	DMA_STR6	DMA1 Stream6 global interrupt	0x0000 0084
adc1_it	25	18	ADC1_2	ADC1 and ADC2 global interrupt	0x0000 0088
adc2_it					
ttfdcan_intr0_it	26	19	FDCAN1_IT0	FDCAN1 Interrupt 0	0x0000 008C
fdcan_intr0_it	27	20	FDCAN2_IT0	FDCAN2 Interrupt 0	0x0000 0090
ttfdcan_intr1_it	28	21	FDCAN1_IT1	FDCAN1 Interrupt 1	0x0000 0094
fdcan_intr1_it	29	22	FDCAN2_IT1	FDCAN2 Interrupt 1	0x0000 0098
exti_exti5_wkup	30	23	EXTI9_5	EXTI Line[9:5] interrupts	0x 0000 009C
exti_exti6_wkup					
exti_exti7_wkup					
exti_exti8_wkup					
exti_exti9_wkup					
tim1_brk_it	31	24	TIM1_BRK	TIM1 break interrupt	0x0000 00A0
tim1_upd_it	32	25	TIM1_UP	TIM1 update interrupt	0x0000 00A4
tim1_trg_it	33	26	TIM1_TRG_COM	TIM1 trigger and commutation interrupts	0x0000 00A8
tim1_cc_it	34	27	TIM_CC	TIM1 capture / compare interrupt	0x0000 00AC
tim2_it	35	28	TIM2	TIM2 global interrupt	0x0000 00B0
tim3_it	36	29	TIM3	TIM3 global interrupt	0x0000 00B4
tim4_it	37	30	TIM4	TIM4 global interrupt	0x0000 00B8

Table 143. NVIC⁽¹⁾ (continued)

Signal	Priority	NVIC position	Acronym	Description	Address offset
i2c1_ev_it	38	31	I2C1_EV	I2C1 event interrupt	0x0000 00BC
exti_i2c1_ev_wkup					
i2c1_err_it	39	32	I2C1_ER	I2C1 error interrupt	0x0000 00C0
i2c2_ev_it	40	33	I2C2_EV	I2C2 event interrupt	0x0000 00C4
exti_i2c2_ev_wkup					
i2c2_err_it	41	34	I2C2_ER	I2C2 error interrupt	0x0000 00C8
spi1_it	42	35	SPI1	SPI1 global interrupt	0x0000 00CC
exti_spi1_it					
spi2_it	43	36	SPI2	SPI2 global interrupt	0x0000 00D0
exti_spi2_it					
usart1_gbl_it	44	37	USART1	USART1 global interrupt	0x0000 00D4
exti_usart1_wkup					
usart2_gbl_it	45	38	USART2	USART2 global interrupt	0x0000 00D8
exti_usart2_wkup					
usart3_gbl_it	46	39	USART3	USART3 global interrupt	0x0000 00DC
exti_usart3_wkup					
exti_exti10_it	47	40	EXTI15_10	EXTI Line[15:10] interrupts	0x0000 00E0
exti_exti11_wkup					
exti_exti12_wkup					
exti_exti13_wkup					
exti_exti14_wkup					
exti_exti15_wkup					
exti_rtc_al	48	41	RTC_ALARM	RTC alarms (A and B) through EXTI Line interrupts	0x0000 00E4
-	49	42	-	-	0x0000 00E8
tim8_brk_it	50	43	TIM8_BRK_TIM12	TIM8 break and TIM12 global interrupts	0x0000 00EC
tim12_gbl_it					
tim8_upd_it	51	44	TIM8_UP_TIM13	TIM8 update and TIM13 global interrupts	0x0000 00F0
tim13_gbl_it					
tim8_trg_it	52	45	TIM8_TRG_COM_TIM14	TIM8 trigger /commutation and TIM14 global interrupts	0x0000 00F4
tim14_gbl_it					
tim8_cc_it	53	46	TIM8_CC	TIM8 capture / compare interrupts	0x0000 00F8

Table 143. NVIC⁽¹⁾ (continued)

Signal	Priority	NVIC position	Acronym	Description	Address offset
dma1_it7	54	47	DMA1_STR7	DMA1 Stream7 global interrupt	0x0000 00FC
fmc_gbl_it	55	48	FMC	FMC global interrupt	0x0000 0100
sdmmc_it	56	49	SDMMC1	SDMMC global interrupt	0x0000 0104
tim5_gbl_it	57	50	TIM5	TIM5 global interrupt	0x0000 0108
spi3_it	58	51	SPI3	SPI3 global interrupt	0x0000 010C
exti_spi3_wkup					
uart4_gbl_it	59	52	UART4	UART4 global interrupt	0x0000 0110
exti_uart4_wkup					
uart5_gbl_it	60	53	UART5	UART5 global interrupt	0x0000 0114
exti_uart5_wkup					
tim6_gbl_it	61	54	TIM6_DAC	TIM6 global interrupt	0x0000 0118
dac_unr_it				DAC underrun error interrupt	
tim7_gbl_it	62	55	TIM7	TIM7 global interrupt	0x0000 011C
dma2_it0	63	56	DMA2_STR0	DMA2 Stream0 interrupt	0x0000 0120
dma2_it1	64	57	DMA2_STR1	DMA2 Stream1 interrupt	0x0000 0124
dma2_it2	65	58	DMA2_STR2	DMA2 Stream2 interrupt	0x0000 0128
dma2_it3	66	59	DMA2_STR3	DMA2 Stream3 interrupt	0x0000 012C
dma2_it4	67	60	DMA2_STR4	DMA2 Stream4 interrupt	0x0000 0130
eth_sbd_intr_it	68	61	ETH	Ethernet global interrupt	0x0000 0134
exti_eth_wkup	69	62	ETH_WKUP	Ethernet wakeup through EXTI line interrupt	0x0000 0138
fdcan_cal_it	70	63	FDCAN_CAL	FDCAN calibration interrupts	0x0000 013C
cm7_sev_it	71	64	-	Arm [®] Cortex [®] -M7 Send even interrupt	0x0000 0140
NC	72	65	-	-	0x0000 0144
NC	73	66	-	-	0x0000 0148
NC	74	67	-	-	0x0000 014C
dma2_it5	75	68	DMA2_STR5	DMA2 Stream5 interrupt	0x0000 0150
dma2_it6	76	69	DMA2_STR6	DMA2 Stream6 interrupt	0x0000 0154
dma2_it7	77	70	DMA2_STR7	DMA2 Stream7 interrupt	0x0000 0158
usart6_gbl_it	78	71	USART6	USART6 global interrupt	0x0000 015C
exti_usart6_wkup				USART6 wakeup interrupt	

Table 143. NVIC⁽¹⁾ (continued)

Signal	Priority	NVIC position	Acronym	Description	Address offset
i2c3_ev_it	79	72	I2C3_EV	I2C3 event interrupt	0x0000 0160
exti_i2c3_ev_wkup					
i2c3_err_it	80	73	I2C3_ER	I2C3 error interrupt	0x0000 0164
usb1_out_it	81	74	OTG_HS_EP1_OUT	OTG_HS out global interrupt	0x0000 0168
usb1_in_it	82	75	OTG_HS_EP1_IN	OTG_HS in global interrupt	0x0000 016C
exti_usb1_wkup	83	76	OTG_HS_WKUP	OTG_HS wakeup interrupt	0x0000 0170
usb1_gbl_it	84	77	OTG_HS	OTG_HS global interrupt	0x0000 0174
dcmi_it	85	78	DCMI	DCMI global interrupt	0x0000 0178
cryp_it	86	79	CRYP	CRYP global interrupt	0x0000 017C
hash_rng_it	87	80	HASH_RNG	HASH and RNG global interrupt	0x0000 0180
cpu_fpu_it	88	81	FPU	CPU FPU	0x0000 0184
uart7_gbl_it	89	82	UART7	UART7 global interrupt	0x0000 0188
exti_uart7_wkup					
uart8_gbl_it	90	83	UART8	UART8 global interrupt	0x0000 018C
exti_uart8_wkup					
spi4_it	91	84	SPI4	SPI4 global interrupt	0x0000 0190
exti_spi4_wkup					
spi5_it	92	85	SPI5	SPI5 global interrupt	0x0000 0194
exti_spi5_wkup					
spi6_it	93	86	SPI6	SPI6 global interrupt	0x0000 0198
exti_spi6_wkup					
sai1_it	94	87	SAI1	SAI1 global interrupt	0x0000 019C
ltdc_it	95	88	LTDC	LCD-TFT global interrupt	0x0000 01A0
ltdc_err_it	96	89	LTDC_ER	LCD-TFT error interrupt	0x0000 01A4
dma2d_gbl_it	97	90	DMA2D	DMA2D global interrupt	0x0000 01A8
-	98	91	SAI2	SAI2 global interrupt	0x0000 01AC
quadspi_it	99	92	QUADSPI	QuadSPI global interrupt	0x0000 01B0
lptim1_it	100	93	LPTIM1	LPTIM1 global interrupt	0x0000 01B4
exti_lptim_wkup					
cec_it	101	94	CEC	HDMI-CEC global interrupt	0x0000 01B8
exti_cec_it					
i2c4_ev_it	102	95	I2C4_EV	I2C4 event interrupt	0x0000 01BC
exti_i2c4_ev_it					

Table 143. NVIC⁽¹⁾ (continued)

Signal	Priority	NVIC position	Acronym	Description	Address offset
i2c4_err_it	103	96	I2C4_ER	I2C4 error interrupt	0x0000 01C0
spdifrx_it	104	97	SPDIF	SPDIFRX global interrupt	0x0000 01C4
usb2_out_it	105	98	OTG_FS_EP1_OUT	OTG_FS out global interrupt	0x0000 01C8
usb2_in_it	106	99	OTG_FS_EP1_IN	OTG_FS in global interrupt	0x0000 01CC
exti_usb2_wkup	107	100	OTG_FS_WKUP	OTG_FS wakeup	0x0000 01D0
usb2_gbl_it	108	101	OTG_FS	OTG_FS global interrupt	0x0000 01D4
dmamux1_ovr_it	109	102	DMAMUX1_OV	DMAMUX1 overrun interrupt	0x0000 01D8
hrtim1_mst_it	110	103	HRTIM1_MST	HRTIM1 master timer interrupt	0x0000 01DC
hrtim1_tima_it	111	104	HRTIM1_TIMA	HRTIM1 timer A interrupt	0x0000 01E0
hrtim1_timb_it	112	105	HRTIM1_TIMB	HRTIM1 timer B interrupt	0x0000 01E4
hrtim1_timc_it	113	106	HRTIM1_TIMC	HRTIM1 timer C interrupt	0x0000 01E8
hrtim1_timd_it	114	107	HRTIM1_TIMD	HRTIM1 timer D interrupt	0x0000 01EC
hrtim1_time_it	115	108	HRTIM1_TIME	HRTIM1 timer E interrupt	0x0000 01F0
hrtim1_fault_it	116	109	HRTIM1_FLT	HRTIM1 fault interrupt	0x0000 01F4
dfsdm1_it0	117	110	DFSDM1_FLT0	DFSDM1 filter 0 interrupt	0x0000 01F8
dfsdm1_it1	118	111	DFSDM1_FLT1	DFSDM1 filter 1 interrupt	0x0000 01FC
dfsdm1_it2	119	112	DFSDM1_FLT2	DFSDM1 filter 2 interrupt	0x0000 0200
dfsdm1_it3	120	113	DFSDM1_FLT3	DFSDM1 filter 3 interrupt	0x0000 0204
sai3_gbl_it	121	114	SAI3	SAI3 global interrupt	0x0000 0208
swpmi_gbl_it	122	115	SWPMI1	SWPMI global interrupt	0x0000 020C
exti_swpmi_wup				SWPMI wakeup	
tim15_gbl_it	123	116	TIM15	TIM15 global interrupt	0x0000 0210
tim16_gbl_it	124	117	TIM16	TIM16 global interrupt	0x0000 0214
tim17_gbl_it	125	118	TIM17	TIM17 global interrupt	0x0000 0218
exti_mdios_wkup	126	119	MDIOS_WKUP	MDIOS wakeup	0x0000 021C
mdios_it	127	120	MDIOS	MDIOS global interrupt	0x0000 0220
jpeg_it	128	121	JPEG	JPEG global interrupt	0x0000 0224
mdma_it	129	122	MDMA	MDMA	0x0000 0228
-	130	-	-	-	0x0000 022C
sdmmc2_it	131	124	SDMMC2	SDMMC global interrupt	0x0000 0230
hsem1_it	132	125	HSEM0	HSEM global interrupt 1	0x0000 0234
-	133	-	-	-	0x0000 0238
adc3_it	134	127	ADC3	ADC3 global interrupt	0x0000 023C

Table 143. NVIC⁽¹⁾ (continued)

Signal	Priority	NVIC position	Acronym	Description	Address offset
dmamux2_ovr_it	135	128	DMAMUX2_OVR	DMAMUX2 overrun interrupt	0x0000 0240
bdma_ch0_it	136	129	BDMA_CH0	BDMA channel 0 interrupt	0x0000 0244
bdma_ch1_it	137	130	BDMA_CH1	BDMA channel 1 interrupt	0x0000 0248
bdma_ch2_it	138	131	BDMA_CH2	BDMA channel 2 interrupt	0x0000 024C
bdma_ch3_it	139	132	BDMA_CH3	BDMA channel 3 interrupt	0x0000 0250
bdma_ch4_it	140	133	BDMA_CH4	BDMA channel 4 interrupt	0x0000 0254
bdma_ch5_it	141	134	BDMA_CH5	BDMA channel 5 interrupt	0x0000 0258
bdma_ch6_it	142	135	BDMA_CH6	BDMA channel 6 interrupt	0x0000 025C
bdma_ch7_it	143	136	BDMA_CH7	BDMA channel 7 interrupt	0x0000 0260
comp_gbl_it	144	137	COMP	COMP1 and COMP2 global interrupt	0x0000 0264
exti_comp1_wkup					
exti_comp2_wkup					
lptim2_it	145	138	LPTIM2	LPTIM2 timer interrupt	0x0000 0268
exti_lptim2_wkup					
lptim3_it	146	139	LPTIM3	LPTIM2 timer interrupt	0x0000 026C
exti_lptim3_wkup					
lptim4_it	147	140	LPTIM4	LPTIM2 timer interrupt	0x0000 0270
exti_lptim4_wkup					
lptim5_it	148	141	LPTIM5	LPTIM2 timer interrupt	0x0000 0274
exti_lptim5_wkup					
lpuart_gbl_it	149	142	LPUART	LPUART global interrupt	0x0000 0278
exti_lpuart_rx_it					
exti_lpuart_tx_it					
exti_d1_wwdg1_wkup	150	143	WWDG1_RST	Window Watchdog interrupt	0x0000 027C
crs_it	151	144	CRS	Clock Recovery System global interrupt	0x0000 0280
ramecc1_it	152	145	RAMECC1	ECC diagnostic global interrupt for RAMECC D1	0x0000 0284
ramecc2_it			RAMECC2	ECC diagnostic global interrupt for RAMECC D2	
ramecc3_it			RAMECC3	ECC diagnostic global interrupt for RAMECC D3	
-	153	146	SAI4	SAI4 global interrupt	0x0000 0288
-	154	147	-	-	0x0000 028C
-	155	148	-	-	0x0000 0290

Table 143. NVIC⁽¹⁾ (continued)

Signal	Priority	NVIC position	Acronym	Description	Address offset
exti_wkup1_wkup	156	149	WKUP	WKUP0 to WKUP5 pins	0x0000 0294
exti_wkup2_wkup					
exti_wkup3_wkup					
exti_wkup4_wkup					
exti_wkup5_wkup					
exti_wkup6_wkup					

1. When different signals are connected to the same NVIC interrupt line, they are OR-ed.

20 Extended interrupt and event controller (EXTI)

The Extended Interrupt and event controller (EXTI) manages wakeup through configurable and direct event inputs. It provides wakeup requests to the Power Control, and generates interrupt requests to the CPU NVIC and to the D3 domain DMAMUX2, and events to the CPU event input.

The EXTI wakeup requests allow the system to be woken up from Stop mode, and the CPU to be woken up from CStop mode.

Both the interrupt request and event request generation can also be used in Run modes.

20.1 EXTI main features

The EXTI main features are the following:

- All Event inputs allow the CPU to wakeup and to generate a CPU interrupt and/or CPU event
- Some Event inputs allow the user to wakeup the D3 domain for autonomous Run mode and generate an interrupt to the D3 domain, i.e. the DMAMUX2

The asynchronous event inputs are classified in 2 groups:

- Configurable events (signals from I/Os or peripherals able to generate a pulse), they have the following features:
 - Selectable active trigger edge
 - Interrupt pending status register bit
 - Individual Interrupt and Event generation mask
 - SW trigger possibility
 - Configurable System D3 domain wakeup events have a D3 Pending mask and status register and may have a D3 interrupt signal.
- Direct events (interrupt and wakeup sources from other peripherals, requiring to be cleared in the peripheral), they feature
 - Fixed rising edge active trigger
 - No interrupt pending status register bit in the EXTI (the interrupt pending status is provided by the peripheral generating the event)
 - Individual Interrupt and Event generation mask
 - No SW trigger possibility
 - Direct system D3 domain wakeup events have a D3 Pending mask and status register and may have a D3 interrupt signal

20.2 EXTI block diagram

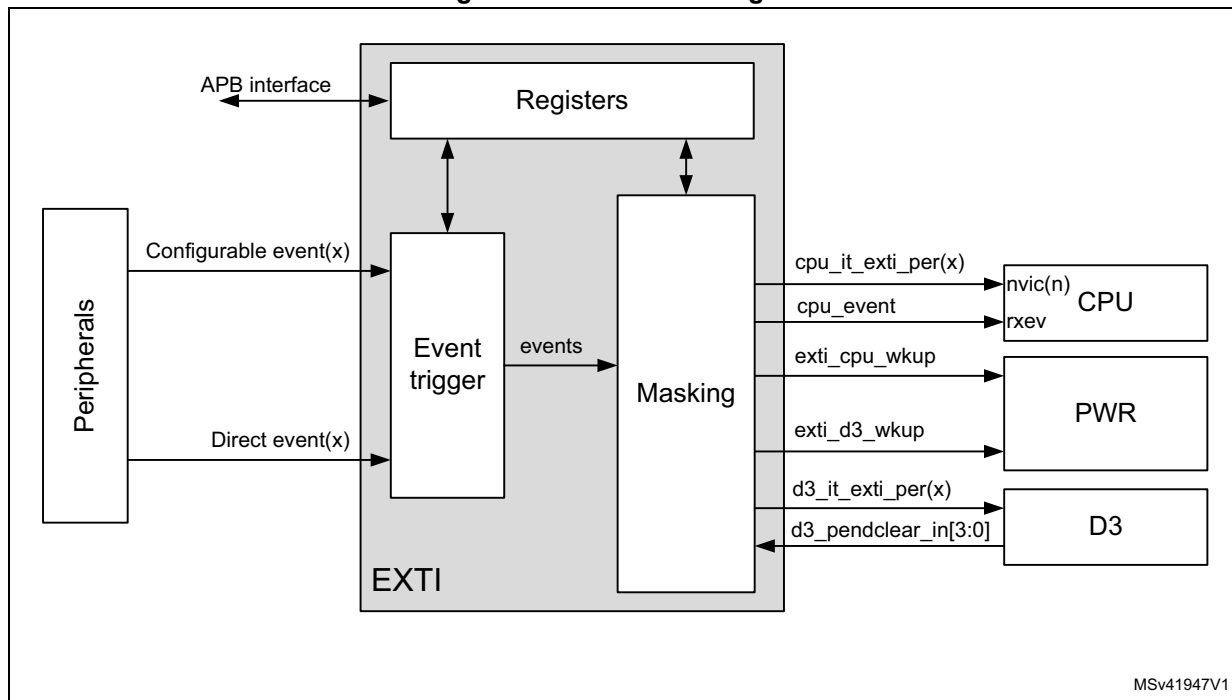
As shown in [Figure 90](#), the EXTI consists of a Register block accessed via an APB interface, an Event input Trigger block, and a Masking block.

The Register block contains all EXTI registers.

The Event input trigger block provides Event input edge triggering logic.

The Masking block provides the Event input distribution to the different wakeup, interrupt and event outputs, and their masking.

Figure 90. EXTI block diagram



20.2.1 EXTI connections between peripherals, CPU, and D3 domain

The peripherals able to generate wakeup events when the system is in Stop mode or the CPU is in CStop mode are connected to an EXTI Configurable event input or Direct Event input:

- Peripheral signals that generate a pulse are connected to an EXTI Configurable Event input. For these events the EXTI provides a CPU status pending bit that has to be cleared.
- Peripheral Interrupt and Wakeup sources that have to be cleared in the peripheral are connected to an EXTI Direct Event input. There is no CPU status pending bit within the EXTI. The Interrupt or Wakeup is cleared by the CPU in the peripheral.

The Event inputs able to wakeup D3 for autonomous Run mode are provided with a D3 domain pending request function, that has to be cleared. This clearing request is taken care of by the signal selected by the Pending clear selection.

The CPU interrupts are connected to their respective CPU NVIC, and, similarly, the CPU event is connected to the CPU rxev input.

The EXTI Wakeup signals are connected to the PWR block, and are used to wakeup the D3 domain and/or the CPU.

The D3 domain interrupts allow the system to trigger events for D3 domain autonomous Run mode operation.

20.3 EXTI functional description

Depending on the EXTI Event input type and wakeup target(s), different logic implementations are used. The applicable features are controlled from register bits:

- Active trigger edge enable, by *EXTI rising trigger selection register (EXTI_RTISR1)*, *EXTI rising trigger selection register (EXTI_RTISR2)*, *EXTI rising trigger selection register (EXTI_RTISR3)*, and *EXTI falling trigger selection register (EXTI_FTISR1)*, *EXTI falling trigger selection register (EXTI_FTISR2)*, *EXTI falling trigger selection register (EXTI_FTISR3)*
- Software trigger, by *EXTI software interrupt event register (EXTI_SWIER1)*, *EXTI software interrupt event register (EXTI_SWIER2)*, *EXTI software interrupt event register (EXTI_SWIER3)*
- CPU Interrupt enable, by *EXTI interrupt mask register (EXTI_CPUIMR1)*, *EXTI interrupt mask register (EXTI_CPUIMR2)*, *EXTI interrupt mask register (EXTI_CPUIMR3)*
- CPU Event enable, by *EXTI event mask register (EXTI_CPUEMR1)*, *EXTI event mask register (EXTI_CPUEMR2)*, *EXTI event mask register (EXTI_CPUEMR3)*
- D3 domain wakeup pending, by *EXTI D3 pending mask register (EXTI_D3PMR1)*, *EXTI D3 pending mask register (EXTI_D3PMR2)*, *EXTI D3 pending mask register (EXTI_D3PMR3)*

Table 144. EXTI Event input configurations and register control⁽¹⁾

Event input type	Wakeup target(s)	Logic implementation	EXTI_RTISR	EXTI_FTISR	EXTI_SWIER	EXTI_CPUIMR	EXTI_CPUEMR	EXTI_D3PMR
Configurable	CPU	Configurable event input, CPU wakeup logic	X	X	X	X	X	-
	Any ⁽²⁾	Configurable event input, Any wakeup logic						X
Direct	CPU	Direct event input, CPU wakeup logic	-	-	-	X	X	-
	Any ⁽²⁾	Direct event input, Any wakeup logic						X

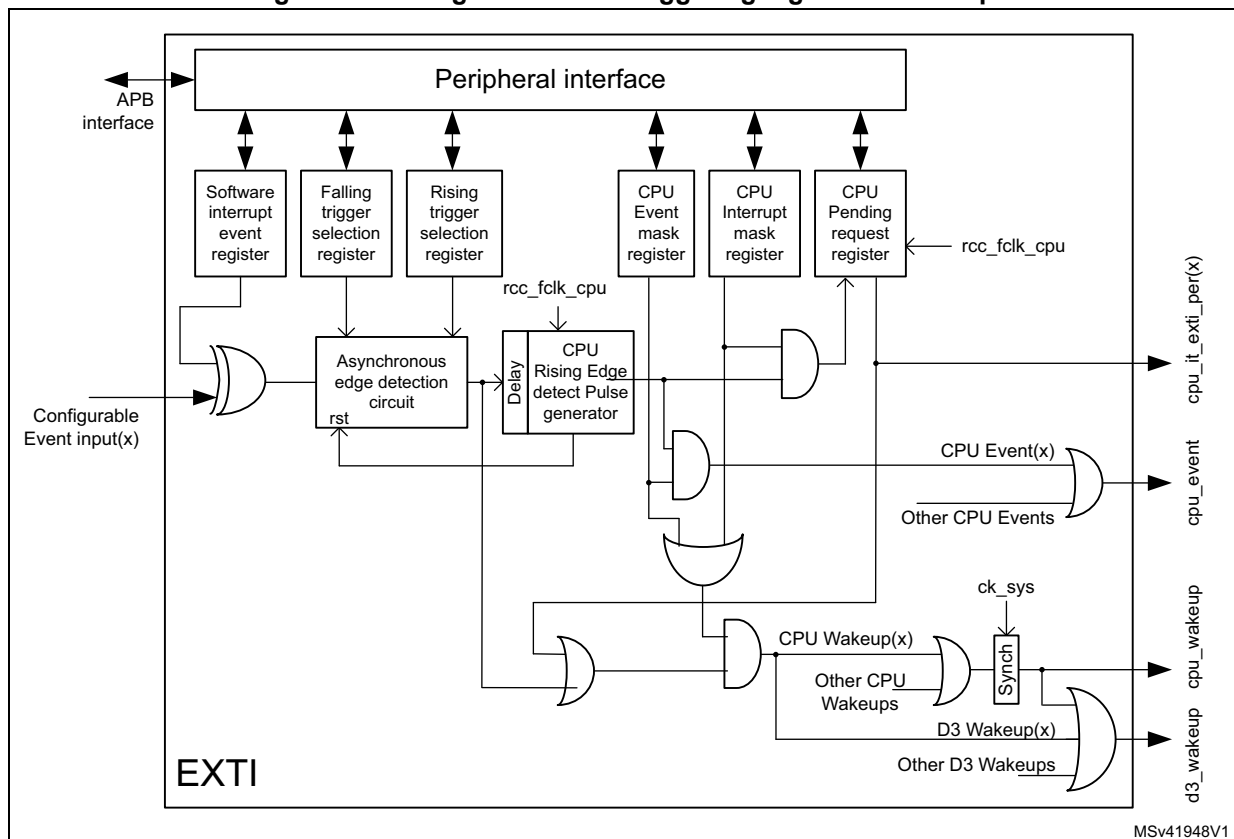
1. X indicates that functionality is available.

2. Waking-up D3 domain for autonomous Run mode, and/or CPU.

20.3.1 EXTI Configurable event input CPU wakeup

Figure 92 is a detailed representation of the logic associated with Configurable Event inputs which will always wake up the CPU.

Figure 91. Configurable event triggering logic CPU wakeup



The Software interrupt event register allows the system to trigger Configurable events by software, writing the *EXTI software interrupt event register (EXTI_SWIER1)*, the *EXTI software interrupt event register (EXTI_SWIER2)*, or the *EXTI software interrupt event register (EXTI_SWIER3)* register bit.

The rising edge *EXTI rising trigger selection register (EXTI_RTSR1)*, *EXTI rising trigger selection register (EXTI_RTSR2)*, *EXTI rising trigger selection register (EXTI_RTSR3)*, and falling edge *EXTI falling trigger selection register (EXTI_FTSR1)*, *EXTI falling trigger selection register (EXTI_FTSR2)*, *EXTI falling trigger selection register (EXTI_FTSR3)* selection registers allow the system to enable and select the Configurable event active trigger edge or both edges.

The devices feature dedicated interrupt mask registers, namely *EXTI interrupt mask register (EXTI_CPUIMR1)* and *EXTI interrupt mask register (EXTI_CPUIMR2)*, *EXTI interrupt mask register (EXTI_CPUIMR3)*, and *EXTI pending register (EXTI_CPUPR1)*, *EXTI pending register (EXTI_CPUPR2)*, *EXTI pending register (EXTI_CPUPR3)* for Configurable events pending request registers. The CPU pending register will only be set for an unmasked CPU interrupt. Each event provides a individual CPU interrupt to the CPU NVIC. The Configurable events interrupts need to be acknowledged by software in the EXTI_CPUPR register.

The devices feature dedicated event mask registers, i.e. *EXTI event mask register (EXTI_C1EMR1)*, *EXTI event mask register (EXTI_C1EMR2)*, and *EXTI event mask register (EXTI_C1EMR3)*. The enabled event then generates an event on the CPU. All events for a CPU are OR-ed together into a single CPU event signal. The CPU Pending register (EXTI_C1UPR) will not be set for an unmasked CPU event.

When a CPU interrupt or CPU event is enabled, the Asynchronous edge detection circuit is reset by the clocked Delay and Rising edge detect pulse generator. This guarantees that the CPU clock is woken up before the Asynchronous edge detection circuit is reset.

Note: A detected Configurable event, enabled by the CPU, is only cleared when the CPU wakes up.

20.3.2 EXTI configurable event input Any wakeup

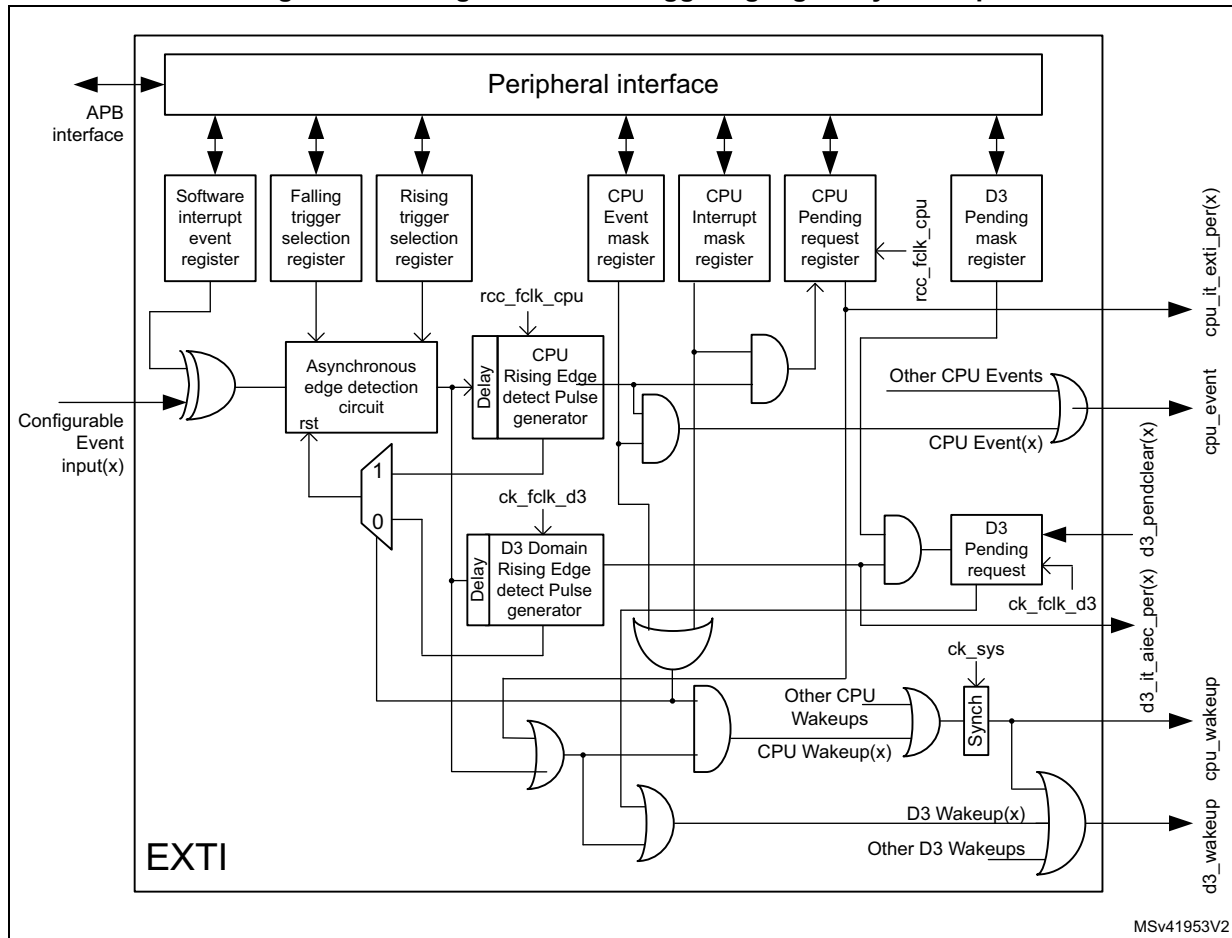
Figure 92 is a detailed representation of the logic associated with Configurable Event inputs that can wakeup D3 domain for autonomous Run mode and/or CPU ("Any" target). It provides the same functionality as the Configurable event input CPU wakeup, with additional functionality to wake up the D3 domain independently.

When all CPU interrupts and CPU events are disabled, the Asynchronous edge detection circuit is reset by the D3 domain clocked Delay and Rising edge detect pulse generator. This guarantees that the D3 domain clock is woken up before the Asynchronous edge detection circuit is reset.

Table 145. Configurable Event input Asynchronous Edge detector reset

EXTI_C1IMR	EXTI_C1EMR	Asynchronous Edge detector reset by
Both = 0		D3 domain clock rising edge detect pulse generator
At least one = 1		CPU clock rising edge detect pulse generator

Figure 92. Configurable event triggering logic Any wakeup



MSv41953V2

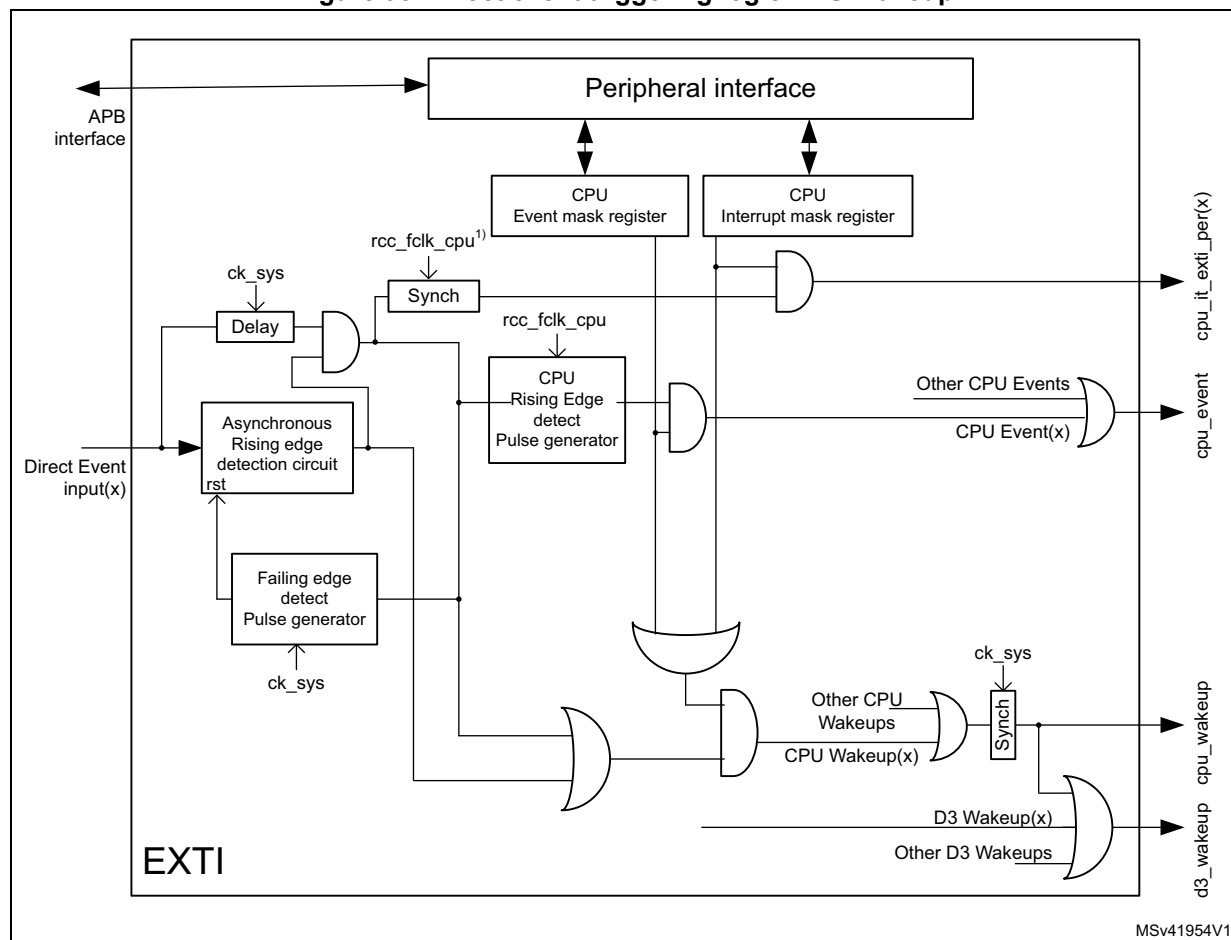
The event triggering logic for “Any” target has additional D3 Pending mask register [EXTI D3 pending mask register \(EXTI_D3PMR1\)](#), [EXTI D3 pending mask register \(EXTI_D3PMR2\)](#), [EXTI D3 pending mask register \(EXTI_D3PMR3\)](#) and D3 Pending request logic. The D3 Pending request logic will only be set for unmasked D3 Pending events. The D3 Pending request logic keeps the D3 domain in Run mode until the D3 Pending request logic is cleared by the selected D3 domain pendclear source.

20.3.3 EXTI direct event input CPU wakeup

Figure 93 is a detailed representation of the logic associated with Direct Event inputs waking up the CPU.

Direct events only provide CPU interrupt enable and CPU event enable functionality.

Figure 93. Direct event triggering logic CPU Wakeup

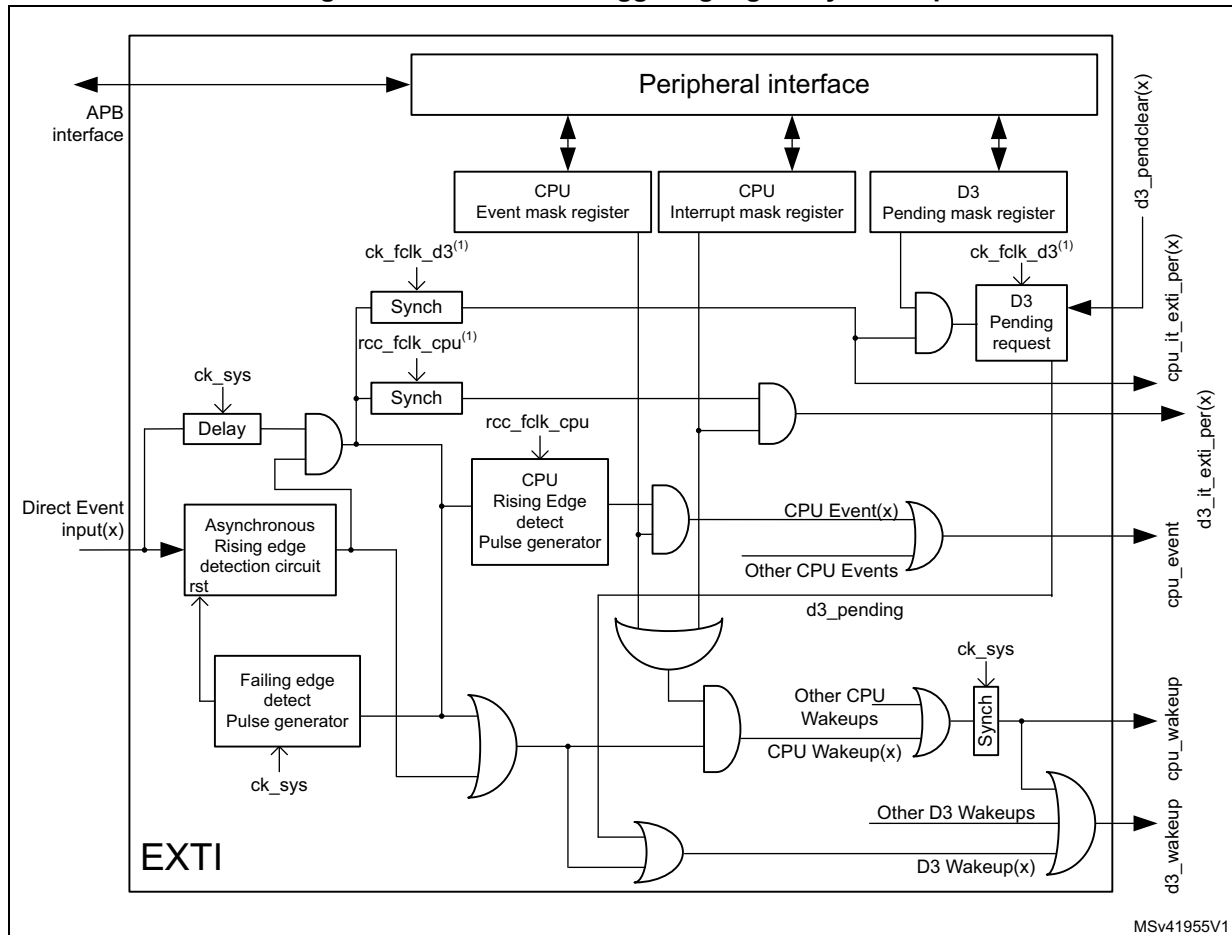


1. The CPU interrupt for asynchronous Direct Event inputs (peripheral Wakeup signals) is synchronized with the CPU clock. The synchronous Direct Event inputs (peripheral interrupt signals), after the asynchronous edge detection, are directly sent to the CPU interrupt without resynchronization.

20.3.4 EXTI direct event input Any wakeup

Figure 94 is a detailed representation of the logic associated with Direct Event inputs waking up D3 domain for autonomous Run mode and/or CPU, (“Any” target). It provides the same functionality as the Direct event input CPU wakeup, plus additional functionality to wakeup the D3 domain independently.

Figure 94. Direct event triggering logic Any Wakeup



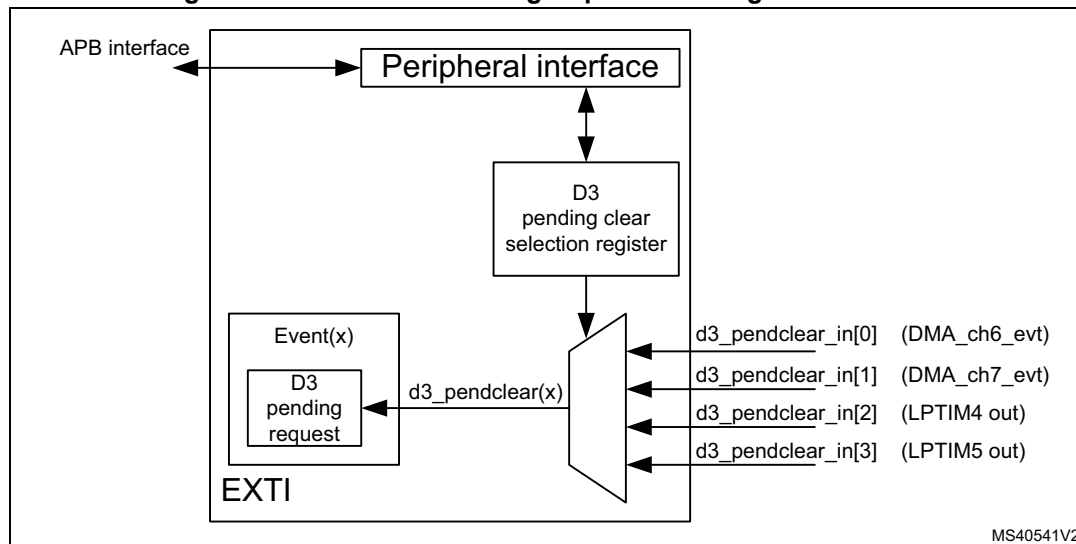
1. The CPU interrupt and D3 domain interrupt for asynchronous Direct Event inputs (peripheral Wakeup signals) are synchronized, respectively, with the CPU clock and the D3 domain clock. The synchronous Direct Event inputs (peripheral interrupt signals), after the asynchronous edge detection, are directly sent to the CPU interrupt and the D3 domain interrupt without resynchronization in the EXTI.

20.3.5 EXTI D3 pending request clear selection

Event inputs able to wake up D3 domain for autonomous Run mode have D3 Pending request logic that can be cleared by the selected D3 pendclear source. For each D3 Pending request a D3 domain pendclear source can be selected from four different inputs.

[Figure 95](#) is a detailed representation of the logic selecting the D3 pendclear source.

Figure 95. D3 domain Pending request clear logic



The D3 Pending request clear selection registers [EXTI D3 pending clear selection register low \(EXTI_D3PCR1L\)](#), [EXTI D3 pending clear selection register high \(EXTI_D3PCR1H\)](#), [EXTI D3 pending clear selection register low \(EXTI_D3PCR2L\)](#), [EXTI D3 pending clear selection register high \(EXTI_D3PCR2H\)](#), [EXTI D3 pending clear selection register low \(EXTI_D3PCR3L\)](#) and [EXTI D3 pending clear selection register high \(EXTI_D3PCR3H\)](#) allow the system to select the source to reset the D3 Pending request.

20.4 EXTI event input mapping

For the sixteen GPIO Event inputs the associated IOPORT pin has to be selected in the SYSCFG register SYSCFG_EXTICRn. The same pin from each IOPORT maps to the corresponding EXTI Event input.

The wakeup capabilities of each Event input are detailed in [Table 146](#). An Event input can either wake up the CPU, and in the case of “Any” can also wake up D3 domain for autonomous Run mode.

The EXTI Event inputs with a connection to the CPU NVIC are indicated in the *Connection to NVIC* column. For the EXTI events not having a connection to the NVIC, the peripheral interrupt is directly connected to the NVIC in parallel with the connection to the EXTI.

All EXTI Event inputs are OR-ed together and connected to the CPU event input (rxev).

Table 146. EXTI Event input mapping

Event input	Source	Event input type	Wakeup target(s)	Connection to NVIC
0 - 15	EXTI[15:0]	Configurable	Any	Yes
16	PVD and AVD ⁽¹⁾	Configurable	CPU only	Yes
17	RTC alarms	Configurable	CPU only	Yes
18	RTC tamper, RTC timestamp, RCC LSECSS ⁽²⁾	Configurable	CPU only	Yes
19	RTC wakeup timer	Configurable	Any	Yes
20	COMP1	Configurable	Any	Yes
21	COMP2	Configurable	Any	Yes
22	I2C1 wakeup	Direct	CPU only	Yes
23	I2C2 wakeup	Direct	CPU only	Yes
24	I2C3 wakeup	Direct	CPU only	Yes
25	I2C4 wakeup	Direct	Any	Yes
26	USART1 wakeup	Direct	CPU only	Yes
27	USART2 wakeup	Direct	CPU only	Yes
28	USART3 wakeup	Direct	CPU only	Yes
29	USART6 wakeup	Direct	CPU only	Yes
30	UART4 wakeup	Direct	CPU only	Yes
31	UART5 wakeup	Direct	CPU only	Yes
32	UART7 wakeup	Direct	CPU only	Yes
33	UART8 wakeup	Direct	CPU only	Yes
34	LPUART1 RX wakeup	Direct	Any	Yes
35	LPUART1 TX wakeup	Direct	Any	Yes
36	SPI1 wakeup	Direct	CPU only	Yes
37	SPI2 wakeup	Direct	CPU only	Yes
38	SPI3 wakeup	Direct	CPU only	Yes
39	SPI4 wakeup	Direct	CPU only	Yes
40	SPI5 wakeup	Direct	CPU only	Yes
41	SPI6 wakeup	Direct	Any	Yes
42	MDIO wakeup	Direct	CPU only	Yes
43	USB1 wakeup	Direct	CPU only	Yes
44	USB2 wakeup	Direct	CPU only	Yes
45	Reserved	-	-	-
46	Reserved	-	-	-
47	LPTIM1 wakeup	Direct	CPU only	Yes
48	LPTIM2 wakeup	Direct	Any	Yes

Table 146. EXTI Event input mapping (continued)

Event input	Source	Event input type	Wakeup target(s)	Connection to NVIC
49	LPTIM2 output	Configurable	Any	No ⁽³⁾
50	LPTIM3 wakeup	Direct	Any	Yes
51	LPTIM3 output	Configurable	Any	No ⁽³⁾
52	LPTIM4 wakeup	Direct	Any	Yes
53	LPTIM5 wakeup	Direct	Any	Yes
54	SWPMI wakeup	Direct	CPU only	Yes
55 ⁽⁴⁾	WKUP1	Direct	CPU only	Yes
56 ⁽⁴⁾	WKUP2	Direct	CPU only	Yes
57 ⁽⁴⁾	WKUP3	Direct	CPU only	Yes
58 ⁽⁴⁾	WKUP4	Direct	CPU only	Yes
59 ⁽⁴⁾	WKUP5	Direct	CPU only	Yes
60 ⁽⁴⁾	WKUP6	Direct	CPU only	Yes
61	RCC interrupt	Direct	CPU only	No ⁽⁵⁾
62	I2C4 Event interrupt	Direct	CPU only	No ⁽⁵⁾
63	I2C4 Error interrupt	Direct	CPU only	No ⁽⁵⁾
64	LPUART1 global Interrupt	Direct	CPU only	No ⁽⁵⁾
65	SPI6 interrupt	Direct	CPU only	No ⁽⁵⁾
66	BDMA CH0 interrupt	Direct	CPU only	No ⁽⁵⁾
67	BDMA CH1 interrupt	Direct	CPU only	No ⁽⁵⁾
68	BDMA CH2 interrupt	Direct	CPU only	No ⁽⁵⁾
69	BDMA CH3 interrupt	Direct	CPU only	No ⁽⁵⁾
70	BDMA CH4 interrupt	Direct	CPU only	No ⁽⁵⁾
71	BDMA CH5 interrupt	Direct	CPU only	No ⁽⁵⁾
72	BDMA CH6 interrupt	Direct	CPU only	No ⁽⁵⁾
73	BDMA CH7 interrupt	Direct	CPU only	No ⁽⁵⁾
74	DMAMUX2 interrupt	Direct	CPU only	No ⁽⁵⁾
75	ADC3 interrupt	Direct	CPU only	No ⁽⁵⁾
76	SAI4 interrupt	Direct	CPU only	No ⁽⁵⁾
77	Reserved	-	-	-
78	Reserved	-	-	-
79	Reserved	-	-	-
80	Reserved	-	-	-
81	Reserved	-	-	-
82	Reserved	-	-	-
83	Reserved	-	-	-

Table 146. EXTI Event input mapping (continued)

Event input	Source	Event input type	Wakeup target(s)	Connection to NVIC
84	Reserved	-	-	-
85	HDMI-CEC wakeup	Configurable	CPU only	Yes
86	ETHERNET wakeup	Configurable	CPU only	Yes
87	HSECSS interrupt	Direct	CPU only	No ⁽⁵⁾
88	Reserved	-	-	-

1. PVD and AVD signals are OR-ed together on the same EXTI event input.
2. RTC Tamper, RTC timestamp and RCC LSECSS signals are OR-ed together on the same EXTI event input.
3. Not available on CPU NVIC, to be used for system wakeup only or CPU event input (rxev).
4. Signals of WKUP1 to WKUP6 correspond to WKUPn pin+1.
5. Available on CPU NVIC directly from the peripheral

20.5 EXTI functional behavior

The Direct event inputs are enabled in the respective peripheral generating the event. The Configurable events are enabled by enabling at least one of the trigger edges.

When in Stop mode an event will always wake up the D3 domain. In system Run and Stop modes an event will always generate an associated D3 domain interrupt. An event will only wake up the CPU when the event associated CPU interrupt is unmasked and/or the CPU event is unmasked.

Table 147. Masking functionality

CPU		Configurable event inputs PRx bits of EXTI_CPUPR	CPU			D3 domain wakeup
Interrupt enable MRx bits of EXTI_CPUIMR	Event enable MRx bits of EXTI_CPUEMR		Interrupt	Event	Wakeup	
0	0	No	Masked	Masked	Masked	Yes ⁽¹⁾ / Masked ⁽²⁾
0	1	No	Masked	Yes	Yes	Yes
1	0	Status latched	Yes	Masked	Yes	Yes
1	1	Status latched	Yes	Yes	Yes	Yes

1. Only for Event inputs that allow the system to wakeup D3 domain for autonomous Run mode (Any target).
2. For Event inputs that will always wake up CPU.

For Configurable event inputs, when the enabled edge(s) occur on the event input, an event request is generated. When the associated CPU interrupt is unmasked, the corresponding pending PRx bit in EXTI_CPUPR is set and the CPU interrupt signal is activated. EXTI_CPUPR PRx pending bit shall be cleared by software writing it to '1'. This will clear the CPU interrupt.

For Direct event inputs, when enabled in the associated peripheral, an event request is generated on the rising edge only. There is no corresponding CPU pending bit. When the associated CPU interrupt is unmasked the corresponding CPU interrupt signal is activated.

The CPU event has to be unmasked to generate an event. When the enabled edge(s) occur on the Event input a CPU event pulse is generated. There is no CPU Event pending bit.

Both a CPU interrupt and a CPU event may be enabled on the same Event input. They will both trigger the same Event input condition(s).

For the Configurable Event inputs an event input request can be generated by software when writing a '1' in the software interrupt/event register EXTI_SWIER.

Whenever an Event input is enabled and a CPU interrupt and/or CPU event is unmasked, the Event input will also generate a D3 domain wakeup next to the CPU wakeup.

Some Event inputs are able to wakeup the D3 domain autonomous Run mode, in this case the CPU interrupt and CPU event are masked, preventing the CPU to be woken up. Two D3 domain autonomous Run mode wakeup mechanisms are supported:

- D3 domain wakeup without pending (EXTI_D3PMR = 0)
 - On a Configurable Event input this mechanism will wake up D3 domain and clear the D3 domain wakeup signal automatically after the Delay + Rising Edge detect Pulse generator.
 - On a Direct Event input this mechanism will wake up D3 domain and clear the D3 domain wakeup signal after the Direct Event input signal is cleared.
- D3 domain wakeup with pending (EXTI_D3PMR = 1)
 - On a Configurable Event input this mechanism will wake up D3 domain and clear the D3 domain wakeup signal after the Delay + Rising Edge detect Pulse generator and when the D3 Pending request is cleared.
 - On a Direct Event input this mechanism will wake up D3 domain and clear the D3 domain wakeup signal after the Direct Event input signal is cleared and when the D3 Pending request is cleared.

20.5.1 EXTI CPU interrupt procedure

- Unmask the Event input interrupt by setting the corresponding mask bits in the EXTI_CPUIMR register.
- For Configurable Event inputs, enable the event input by setting either one or both the corresponding trigger edge enable bits in EXTI_RTSR and EXTI_FTSR registers.
- Enable the associated interrupt source in the CPU NVIC or use the SEVONPEND, so that an interrupt coming from the CPU interrupt signal is detectable by the CPU after a WFI/WFE instruction.
 - For Configurable event inputs the associated EXTI pending bit needs to be cleared.

20.5.2 EXTI CPU event procedure

- Unmask the Event input by setting the corresponding mask bits of the EXTI_CPUEMR register.
- For Configurable Event inputs, enable the event input by setting either one or both the corresponding trigger edge enable bits in EXTI_RTSR and EXTI_FTSR registers.
- The CPU event signal is detected by the CPU after a WFE instruction.
 - For Configurable event inputs there is no EXTI pending bit to clear.

20.5.3 EXTI CPU wakeup procedure

- Unmask the Event input by setting at least one of the corresponding mask bits in the EXTI_CPUIMR and/or EXTI_CPUEMR registers. The CPU wakeup is generated at the same time as the unmasked CPU interrupt and/or CPU event.
- For Configurable Event inputs, enable the event input by setting either one or both the corresponding trigger edge enable bits in EXTI_RTSR and EXTI_FTSR registers.
- Direct Events will automatically generate a CPU wakeup.

20.5.4 EXTI D3 domain wakeup for autonomous Run mode procedure

- Mask the Event input for waking up the CPU, by clearing both the corresponding mask bits in the EXTI_CPUIMR and/or EXTI_CPUEMR registers.
- For Configurable Event inputs, enable the event input by setting either one or both the corresponding trigger edge enable bits in EXTI_RTSR and EXTI_FTSR registers.
- Direct Events will automatically generate a D3 domain wakeup.
- Select the D3 domain wakeup mechanism in EXTI_D3PMR.
 - When D3 domain wakeup without pending (EXTI_PMR = 0) is selected, the Wakeup will be cleared automatically following the clearing of the Event input.
 - When D3 domain wakeup with pending (EXTI_PMR = 1) is selected the Wakeup needs to be cleared by a selected D3 domain pendclear source.
A pending D3 domain wakeup signal can also be cleared by FW clearing the associated EXTI_D3PMR register bit.
- After the D3 domain wakeup a D3 domain interrupt is generated.
 - Configurable Event inputs will generate a pulse on D3 domain interrupt.
 - Direct Event inputs will activate the D3 domain interrupt until the event input is cleared in the peripheral.

20.5.5 EXTI software interrupt/event trigger procedure

Any of the Configurable Event inputs can be triggered from the software interrupt/event register (the associated CPU interrupt and/or CPU event shall be enabled by their respective procedure).

- Enable the Event input by setting at least one of the corresponding edge trigger bits in the EXTI_RTSR and/or EXTI_FTSR registers.
- Unmask the software interrupt/event trigger by setting at least one of the corresponding mask bits in the EXTI_CPUIMR and/or EXTI_CPUEMR registers.
- Trigger the software interrupt/event by writing “1” to the corresponding bit in the EXTI_SWIER register.
- The Event input may be disabled by clearing the EXTI_RTSR and EXTI_FTSR register bits.

Note: An edge on the Configurable event input will also trigger an interrupt/event.

A software trigger can be used to set the D3 Pending request logic, keeping the D3 domain in Run until the D3 Pending request logic is cleared.

20.6 EXTI registers

Every register can only be accessed with 32-bit (word). A byte or half-word cannot be read or written.

20.6.1 EXTI rising trigger selection register (EXTI_RTSR1)

Address offset: 0x00

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR21	TR20	TR19	TR18	TR17	TR16
										r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:22 Reserved, must be kept at reset value.

Bits 21:0 **TRx**: Rising trigger event configuration bit of Configurable Event input x.⁽¹⁾

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

- The Configurable event inputs are edge triggered, no glitch must be generated on these inputs. If a rising edge on the Configurable event input occurs during writing of the register, the associated pending bit will not be set. Rising and falling edge triggers can be set for the same Configurable Event input. In this case, both edges generate a trigger.

20.6.2 EXTI falling trigger selection register (EXTI_FTSR1)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR21	TR20	TR19	TR18	TR17	TR16
										r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:22 Reserved, must be kept at reset value.

Bits 21:0 **TRx**: Falling trigger event configuration bit of Configurable Event input x.⁽¹⁾

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line.

- The Configurable event inputs are edge triggered, no glitch must be generated on these inputs. If a falling edge on the Configurable event input occurs during writing of the register, the associated pending bit will not be set. Rising and falling edge triggers can be set for the same Configurable Event input. In this case, both edges generate a trigger.

20.6.3 EXTI software interrupt event register (EXTI_SWIER1)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWIER 21	SWIER 20	SWIER 19	SWIER 18	SWIER 17	SWIER 16
										rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWIER 15	SWIER 14	SWIER 13	SWIER 12	SWIER 11	SWIER 10	SWIER 9	SWIER 8	SWIER 7	SWIER 6	SWIER 5	SWIER 4	SWIER 3	SWIER 2	SWIER 1	SWIER 0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:22 Reserved, must be kept at reset value.

Bits 21:0 **SWIERx**: Software interrupt on line x

Will always return 0 when read.

0: Writing 0 has no effect.

1: Writing a 1 to this bit will trigger an event on line x. This bit is auto cleared by HW.

20.6.4 EXTI D3 pending mask register (EXTI_D3PMR1)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	MR25	Res.	Res.	Res.	MR21	MR20	MR19	Res.	Res.	Res.
						rw				rw	rw	rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:26 Reserved, must be kept at reset value.

Bit 25 **MRx**: D3 Pending Mask on Event input x

0: D3 Pending request from Line x is masked. Writing this bit to 0 will also clear the D3 Pending request.

1: D3 Pending request from Line x is unmasked. The D3 domain pending signal when triggered will keep D3 domain wakeup active until cleared.

Bits 24:22 Reserved, must be kept at reset value.

Bits 21:19 **MRx**: D3 Pending Mask on Event input x

0: D3 Pending request from Line x is masked. Writing this bit to 0 will also clear the D3 Pending request.

1: D3 Pending request from Line x is unmasked. The D3 domain pending signal when triggered will keep D3 domain wakeup active until cleared.

Bits 18:16 Reserved, must be kept at reset value.

Bits 15:0 **MRx**: D3 Pending Mask on Event input x

0: D3 Pending request from Line x is masked. Writing this bit to 0 will also clear the D3 Pending request.

1: D3 Pending request from Line x is unmasked. The D3 domain pending signal when triggered will keep D3 domain wakeup active until cleared.

20.6.5 EXTI D3 pending clear selection register low (EXTI_D3PCR1L)

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PCS15		PCS14		PCS13		PCS12		PCS11		PCS10		PCS9		PCS8	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PCS7		PCS6		PCS5		PCS4		PCS3		PCS2		PCS1		PCS0	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **PCSx**: D3 Pending request clear input signal selection on Event input x = truncate (n/2)

00: DMA ch6 event selected as D3 domain pendclear source

01: DMA ch7 event selected as D3 domain pendclear source

10: LPTIM4 out selected as D3 domain pendclear source

11: LPTIM5 out selected as D3 domain pendclear source

20.6.6 EXTI D3 pending clear selection register high (EXTI_D3PCR1H)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCS25		Res.	Res.
												rw	rw		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	PCS21		PCS20		PCS19		Res.	Res.	Res.	Res.	Res.	Res.
				rw	rw	rw	rw	rw	rw						

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:18 **PCSx**: D3 Pending request clear input signal selection on Event input x = truncate ((n+32)/2)

00: DMA ch6 event selected as D3 domain pendclear source

01: DMA ch7 event selected as D3 domain pendclear source

10: LPTIM4 out selected as D3 domain pendclear source

11: LPTIM5 out selected as D3 domain pendclear source

Bits 17:12 Reserved, must be kept at reset value.

Bits 11:6 **PCSx**: D3 Pending request clear input signal selection on Event input x = truncate ((n+32)/2)

00: DMA ch6 event selected as D3 domain pendclear source

01: DMA ch7 event selected as D3 domain pendclear source

10: LPTIM4 out selected as D3 domain pendclear source

11: LPTIM5 out selected as D3 domain pendclear source

Bits 5:0 Reserved, must be kept at reset value.

20.6.7 EXTI rising trigger selection register (EXTI_RTSTR2)

Address offset: 0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR51	Res.	TR49	Res.
												rw		rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:20 Reserved, must be kept at reset value.

Bit 19 **TRx**: Rising trigger event configuration bit of Configurable Event input x+32.⁽¹⁾

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Bit 18 Reserved, must be kept at reset value.

Bit 17 **TRx**: Rising trigger event configuration bit of Configurable Event input x+32.⁽¹⁾

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Bits 16:0 Reserved, must be kept at reset value.

- The Configurable event inputs are edge triggered, no glitch must be generated on these inputs.
If a rising edge on the Configurable event input occurs during writing of the register, the associated pending bit will not be set.
Rising and falling edge triggers can be set for the same Configurable Event input. In this case, both edges generate a trigger.

20.6.8 EXTI falling trigger selection register (EXTI_FTSR2)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR51	Res.	TR49	Res.
												rw		rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:20 Reserved, must be kept at reset value.

Bit 19 **TRx**: Falling trigger event configuration bit of Configurable Event input x+32.⁽¹⁾

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line

Bit 18 Reserved, must be kept at reset value.

Bit 17 **TRx**: Falling trigger event configuration bit of Configurable Event input x+32.⁽¹⁾

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line

Bits 16:0 Reserved, must be kept at reset value.

1. The Configurable event inputs are edge triggered, no glitch must be generated on these inputs.
If a falling edge on the Configurable event input occurs during writing of the register, the associated pending bit will not be set.
Rising and falling edge triggers can be set for the same Configurable Event input. In this case, both edges generate a trigger.

20.6.9 EXTI software interrupt event register (EXTI_SWIER2)

Address offset: 0x28

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWIER 51	Res.	SWIER 49	Res.
												rw		rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:20 Reserved, must be kept at reset value.

Bit 19 **SWIERx**: Software interrupt on line x+32

Will always return 0 when read.

0: Writing 0 has no effect.

1: Writing a 1 to this bit will trigger an event on line x. This bit is auto cleared by HW.

Bit 18 Reserved, must be kept at reset value.

Bit 17 **SWIERx**: Software interrupt on line x+32
Will always return 0 when read.

0: Writing 0 has no effect.

1: Writing a 1 to this bit will trigger an event on line x. This bit is auto cleared by HW.

Bits 16:0 Reserved, must be kept at reset value.

20.6.10 EXTI D3 pending mask register (EXTI_D3PMR2)

Address offset: 0x2C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR53	MR52	MR51	MR50	MR49	MR48
										rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	MR41	Res.	Res.	Res.	Res.	Res.	MR35	MR34	Res.	Res.
						rw						rw	rw		

Bits 31:22 Reserved, must be kept at reset value.

Bits 21:16 **MRx**: D3 Pending Mask on Event input x+32

0: D3 Pending request from Line x+32 is masked. Writing this bit to 0 will also clear the D3 Pending request.

1: D3 Pending request from Line x+32 is unmasked. The D3 domain pending signal when triggered will keep D3 domain wakeup active until cleared.

Bits 15:10 Reserved, must be kept at reset value.

Bit 9 **MRx**: D3 Pending Mask on Event input x+32

0: D3 Pending request from Line x+32 is masked. Writing this bit to 0 will also clear the D3 Pending request.

1: D3 Pending request from Line x+32 is unmasked. The D3 domain pending signal when triggered will keep D3 domain wakeup active until cleared.

Bits 8:4 Reserved, must be kept at reset value.

Bits 3:2 **MRx**: D3 Pending Mask on Event input x+32

0: D3 Pending request from Line x+32 is masked. Writing this bit to 0 will also clear the D3 Pending request.

1: D3 Pending request from Line x+32 is unmasked. The D3 domain pending signal when triggered will keep D3 domain wakeup active until cleared.

Bits 1:0 Reserved, must be kept at reset value.

20.6.11 EXTI D3 pending clear selection register low (EXTI_D3PCR2L)

Address offset: 0x30

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCS41		Res.	Res.
												rw	rw		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCS35		PCS34		Res.	Res.	Res.	Res.
								rw	rw	rw	rw				

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:18 **PCSx**: D3 Pending request clear input signal selection on Event input x = truncate ((n+64)/2)

00: DMA ch6 event selected as D3 domain pendclear source

01: DMA ch7 event selected as D3 domain pendclear source

10: LPTIM4 out selected as D3 domain pendclear source

11: LPTIM5 out selected as D3 domain pendclear source

Bits 17:8 Reserved, must be kept at reset value.

Bits 7:4 **PCSx**: D3 Pending request clear input signal selection on Event input x= truncate ((n+64)/2)

00: DMA ch6 event selected as D3 domain pendclear source

01: DMA ch7 event selected as D3 domain pendclear source

10: LPTIM4 out selected as D3 domain pendclear source

11: LPTIM5 out selected as D3 domain pendclear source

Bits 3:0 Reserved, must be kept at reset value.

20.6.12 EXTI D3 pending clear selection register high (EXTI_D3PCR2H)

Address offset: 0x34

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	PCS53		PCS52		PCS51		PCS50		PCS49		PCS48	
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **PCSx**: D3 Pending request clear input signal selection on Event input x= truncate ((n+96)/2)

00: DMA ch6 event selected as D3 domain pendclear source

01: DMA ch7 event selected as D3 domain pendclear source

10: LPTIM4 out selected as D3 domain pendclear source

11: LPTIM5 out selected as D3 domain pendclear source

20.6.13 EXTI rising trigger selection register (EXTI_RTSTR3)

Address offset: 0x40

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR86	TR85	TR84	Res.	TR82	Res.	Res.
									rw	rw	rw		rw		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:20 **TRx**: Rising trigger event configuration bit of Configurable Event input x+64.⁽¹⁾

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Bit 19 Reserved, must be kept at reset value.

Bit 18 **TRx**: Rising trigger event configuration bit of Configurable Event input x+64.⁽¹⁾

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Bits 17:0 Reserved, must be kept at reset value.

1. The Configurable event inputs are edge triggered, no glitch must be generated on these inputs.
If a rising edge on the Configurable event input occurs during writing of the register, the associated pending bit will not be set.
Rising and falling edge triggers can be set for the same Configurable Event input. In this case, both edges generate a trigger.

20.6.14 EXTI falling trigger selection register (EXTI_FTSR3)

Address offset: 0x44

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR86	TR85	TR84	Res.	TR82	Res.	Res.
									rw	rw	rw		rw		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:20 **TRx**: Falling trigger event configuration bit of Configurable Event input x+64.⁽¹⁾

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line

Bit 19 Reserved, must be kept at reset value.

Bit 18 **TRx**: Falling trigger event configuration bit of Configurable Event input x+64.⁽¹⁾

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line

Bits 17:0 Reserved, must be kept at reset value.

1. The Configurable event inputs are edge triggered, no glitch must be generated on these inputs. If a falling edge on the Configurable event input occurs during writing of the register, the associated pending bit will not be set. Rising and falling edge triggers can be set for the same Configurable Event input. In this case, both edges generate a trigger.

20.6.15 EXTI software interrupt event register (EXTI_SWIER3)

Address offset: 0x48

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWIER 86	SWIER 85	SWIER 84	Res.	SWIER 82	Res.	Res.
									rw	rw	rw		rw		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:20 **SWIERx**: Software interrupt on line x+64

Will always return 0 when read.

0: Writing 0 has no effect.

1: Writing a 1 to this bit will trigger an event on line x. This bit is auto cleared by HW.

Bit 19 Reserved, must be kept at reset value.

Bit 18 **SWIERx**: Software interrupt on line x+64

Will always return 0 when read.

0: Writing 0 has no effect.

1: Writing a 1 to this bit will trigger an event on line x. This bit is auto cleared by HW.

Bits 17:0 Reserved, must be kept at reset value.

20.6.16 EXTI D3 pending mask register (EXTI_D3PMR3)

Address offset: 0x4C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR88	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
							rw								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:25 Reserved, must be kept at reset value.

Bit 24 **MRx**: D3 Pending Mask on Event input x+64

0: D3 Pending request from Line x+64 is masked. Writing this bit to 0 will also clear the D3 Pending request.

1: D3 Pending request from Line x+64 is unmasked. The D3 domain pending signal when triggered will keep D3 domain wakeup active until cleared.

Bits 23:0 Reserved, must be kept at reset value.

20.6.17 EXTI D3 pending clear selection register low (EXTI_D3PCR3L)

Address offset: 0x50

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:0 Reserved, must be kept at reset value.

20.6.18 EXTI D3 pending clear selection register high (EXTI_D3PCR3H)

Address offset: 0x54

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCS88	
														rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:18 Reserved, must be kept at reset value.

Bits 17:16 **PCSx**: D3 Pending request clear input signal selection on Event input x= truncate ((n+160)/2)

00: DMA ch6 event selected as D3 domain pendclear source

01: DMA ch7 event selected as D3 domain pendclear source

10: LPTIM4 out selected as D3 domain pendclear source

11: LPTIM5 out selected as D3 domain pendclear source

Bits 15:0 Reserved, must be kept at reset value.

20.6.19 EXTI interrupt mask register (EXTI_CPUIMR1)

Address offset: 0x80

Reset value: 0xFFC0 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MR31	MR30	MR29	MR28	MR27	MR26	MR25	MR24	MR23	MR22	MR21	MR20	MR19	MR18	MR17	MR16
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:22 **MRx**: CPU interrupt Mask on Direct Event input x⁽¹⁾

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is unmasked

Bits 21:0 **MRx**: CPU interrupt Mask on Configurable Event input x⁽²⁾

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is unmasked

1. The reset value for Direct Event inputs is set to '1' in order to enable the interrupt by default.
2. The reset value for Configurable Event inputs is set to '0' in order to disable the interrupt by default.

20.6.20 EXTI event mask register (EXTI_CPUEMR1)

Address offset: 0x84

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MR31	MR30	MR29	MR28	MR27	MR26	MR25	MR24	MR23	MR22	MR21	MR20	MR19	MR18	MR17	MR16
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **MRx**: CPU Event mask on Event input x

0: Event request from Line x is masked

1: Event request from Line x is unmasked

20.6.21 EXTI pending register (EXTI_CPUPR1)

Address offset: 0x88

Reset value: undefined

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PR21	PR21	PR19	PR18	PR17	PR16
										rc1	rc1	rc1	rc1	rc1	rc1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PR15	PR14	PR13	PR12	PR11	PR10	PR9	PR8	PR7	PR6	PR5	PR4	PR3	PR2	PR1	PR0
rc1	rc1	rc1	rc1	rc1	rc1	rc1	rc1	rc1	rc1	rc1	rc1	rc1	rc1	rc1	rc1

Bits 31:22 Reserved, must be kept at reset value.

Bits 21:0 **PRx**: Configurable event inputs x Pending bit

0: No trigger request occurred

1: selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line. This bit is cleared by writing a 1 into the bit or by changing the sensitivity of the edge detector.

20.6.22 EXTI interrupt mask register (EXTI_CPUIMR2)

Address offset: 0x90

Reset value: 0xFFFF FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MR63	MR62	MR61	MR60	MR59	MR58	MR57	MR56	MR55	MR54	MR53	MR52	MR51	MR50	MR49	MR48
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR47	MR46	Res.	MR44	MR43	MR42	MR41	MR40	MR39	MR38	MR37	MR36	MR35	MR34	MR33	MR32
rw	rw	1	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:20 **MRx**: CPU Interrupt Mask on Direct Event input x+32⁽¹⁾

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is unmasked

Bit 19 **MRx**: CPU interrupt Mask on Configurable Event input x+32⁽²⁾

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is unmasked

Bit 18 **MRx**: CPU Interrupt Mask on Direct Event input x+32⁽¹⁾

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is unmasked

Bit 17 **MRx**: CPU interrupt Mask on Configurable Event input x+32⁽²⁾

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is unmasked

Bits 16:14 **MRx**: CPU Interrupt Mask on Direct Event input x+32 ⁽¹⁾

- 0: Interrupt request from Line x is masked
- 1: Interrupt request from Line x is unmasked

Bit 13 Reserved, must be kept at reset value (1).

Bits 12:0 **MRx**: CPU Interrupt Mask on Direct Event input x+32 ⁽¹⁾

- 0: Interrupt request from Line x is masked
- 1: Interrupt request from Line x is unmasked

1. The reset value for Direct Event inputs is set to '1' in order to enable the interrupt by default.
2. The reset value for Configurable Event inputs is set to '0' in order to disable the interrupt by default.

20.6.23 EXTI event mask register (EXTI_CPUEMR2)

Address offset: 0x94

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MR63	MR62	MR61	MR60	MR59	MR58	MR57	MR56	MR55	MR54	MR53	MR52	MR51	MR50	MR49	MR48
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR47	MR46	Res.	MR44	MR43	MR42	MR41	MR40	MR39	MR38	MR37	MR36	MR35	MR34	MR33	MR32
rw	rw	0	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:14 **MRx**: CPU Event mask on Event input x+32

- 0: Event request from Line x is masked
- 1: Event request from Line x is unmasked

Bit 13 Reserved, must be kept at reset value.

Bits 12:0 **MRx**: CPU Event mask on Event input x+32

- 0: Event request from Line x is masked
- 1: Event request from Line x is unmasked

20.6.24 EXTI pending register (EXTI_CPUPR2)

Address offset: 0x98

Reset value: undefined

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PR51	Res.	PR49	Res.
												rc1		rc1	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:20 Reserved, must be kept at reset value.

Bit 19 **PRx**: Configurable event inputs x+32 Pending bit

0: No trigger request occurred

1: selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line. This bit is cleared by writing a 1 into the bit or by changing the sensitivity of the edge detector.

Bit 18 Reserved, must be kept at reset value.

Bit 17 **PRx**: Configurable event inputs x+32 Pending bit

0: No trigger request occurred

1: selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line. This bit is cleared by writing a 1 into the bit or by changing the sensitivity of the edge detector.

Bits 16:0 Reserved, must be kept at reset value.

20.6.25 EXTI interrupt mask register (EXTI_CPUIMR3)

Address offset: 0xA0

Reset value: 0x018B FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res	Res	Res	Res	Res	Res	Res	MR88	MR87	MR86	MR85	MR84	Res	MR82	Res	MR80
							r/w	r/w	r/w	r/w	r/w		r/w		r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR79	MR78	MR77	MR76	MR75	MR74	MR73	MR72	MR71	MR70	MR69	MR68	MR67	MR66	MR65	MR64
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:25 Reserved, must be kept at reset value.

Bits 24:23 **MRx**: CPU Interrupt Mask on Direct Event input x+64 ⁽¹⁾

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is unmasked

Bits 22:20 **MRx**: CPU interrupt Mask on Configurable Event input x+64 ⁽²⁾

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is unmasked

Bit 19 Reserved, must be kept at reset value (1).

Bit 18 **MRx**: CPU interrupt Mask on Configurable Event input x+64 ⁽²⁾

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is unmasked

Bit 17 Reserved, must be kept at reset value (1).

Bits 16:0 **MRx**: CPU Interrupt Mask on Direct Event input x+64 ⁽¹⁾

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is unmasked

1. The reset value for Direct Event inputs is set to '1' in order to enable the interrupt by default.

2. The reset value for Configurable Event inputs is set to '0' in order to disable the interrupt by default.

20.6.26 EXTI event mask register (EXTI_CPUEMR3)

Address offset: 0xA4

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR88	MR87	MR86	MR85	MR84	Res.	MR82	Res.	MR80
							rw	rw	rw	rw	rw		rw		rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR79	MR78	MR77	MR76	MR75	MR74	MR73	MR72	MR71	MR70	MR69	MR68	MR67	MR66	MR65	MR64
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bits 24:0 **MRx**: CPU Event mask on Event input x+64

0: Event request from Line x is masked

1: Event request from Line x is unmasked

20.6.27 EXTI pending register (EXTI_CPUPR3)

Address offset: 0xA8

Reset value: undefined

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PR86	PR85	PR84	Res.	PR82	Res.	Res.
									rc1	rc1	rc1		rc1		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:20 **PRx**: Configurable event inputs x+64 Pending bit

0: No trigger request occurred

1: selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line. This bit is cleared by writing a 1 into the bit or by changing the sensitivity of the edge detector.

Bit 19 Reserved, must be kept at reset value.

Bit 18 **PRx**: Configurable event inputs x+64 Pending bit

0: No trigger request occurred

1: selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line. This bit is cleared by writing a 1 into the bit or by changing the sensitivity of the edge detector.

Bits 17:0 Reserved, must be kept at reset value.

20.6.28 EXTI register map

The following table gives the EXTI register map and the reset values.

Table 148. Asynchronous interrupt/event controller register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	EXTI_RTISR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR[21:0]																					
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x04	EXTI_FTSR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR[21:0]																					
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x08	EXTI_SWIER1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWIER[21:0]																					
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	EXTI_D3PMR1	Res.	Res.	Res.	Res.	Res.	MR[25]	Res.	Res.	Res.	Res.	MR[21:19]				Res.	Res.	Res.	MR[15:0]														
	Reset value						0					0	0	0				0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x10	EXTI_D3PCR1L	PCS[15]			PCS[14]		PCS[13]		PCS[12]		PCS[11]		PCS[10]		PCS[9]		PCS[8]		PCS[7]		PCS[6]		PCS[5]		PCS[4]		PCS[3]		PCS[2]		PCS[1]		PCS[0]
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x14	EXTI_D3PCR1H	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCS[25]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCS[21]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value														0	0						0	0	0	0	0	0						
0x20	EXTI_RTISR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR[51]	Res.	TR[49]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value														0		0																
0x24	EXTI_FTSR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR[51]	Res.	TR[49]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value														0		0																
0x28	EXTI_SWIER2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWIER[51]	Res.	SWIER[49]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value														0		0																
0x2C	EXTI_D3PMR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR[53:48]						Res.	Res.	Res.	Res.	Res.	Res.	MR[41]	Res.	Res.	Res.	Res.	MR[35]	Res.	MR[34]	Res.	Res.
	Reset value											0	0	0	0	0	0						0						0	0	0	0	0
0x30	EXTI_D3PCR2L	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCS[41]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCS[35]	Res.	Res.	PCS[34]	Res.	Res.	Res.	Res.
	Reset value														0	0									0	0	0	0		0	0	0	0
0x34	EXTI_D3PCR2H	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCS[53]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value																						0	0	0	0	0	0	0	0	0	0	0
0x40	EXTI_RTISR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR[86]	TR[85]	TR[84]	Res.	TR[82]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value											0	0	0		0																	

Table 148. Asynchronous interrupt/event controller register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x44	EXTI_FTSR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR[86]	TR[85]	TR[84]	Res.	TR[82]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value										0	0	0	0	0																			
0x48	EXTI_SWIER3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWIER[86]	SWIER[85]	SWIER[84]	Res.	SWIER[82]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value										0	0	0	0	0																			
0x4C	EXTI_D3PMR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR[88]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value								0																									
0x50	EXTI_D3PCR3L	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value																																	
0x54	EXTI_D3PCR3H	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCS[88]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value															0	0																	
0x58-0x7C	Reserved																																	
0x80	EXTI_CPUIMR1	MR[31:22]										MR[21:0]																						
	Reset value	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x84	EXTI_CPUEMR1	MR[31:0]																																
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x88	EXTI_CPUPR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PR[21:0]																						
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x90	EXTI_CPUIMR2	MR[63:52]										MR[51]	MR[50]	MR[49]	MR[48:46]				Res.	MR[44:32]														
	Reset value	1	1	1	1	1	1	1	1	1	1	1	0	1	0	1	1	1			1	1	1	1	1	1	1	1	1	1	1	1	1	1
0x94	EXTI_CPUEMR2	MR[63:46]																		Res.	MR[44:32]													
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	
0x98	EXTI_CPUPR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PR[51]	Res.	PR[49]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value													0		0																		
0xA0	EXTI_CPUIMR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR[88]	MR[87]	MR[86]	MR[85]	MR[84]	Res.	MR[82]	Res.	MR[80:64]																	
	Reset value								1	1	0	0	0	0	0		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
0xA4	EXTI_CPUEMR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR[88:84]				Res.	MR[82]	Res.	MR[80:64]																		
	Reset value								0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 148. Asynchronous interrupt/event controller register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xA8	EXTI_CPUPR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PR[86]	PR[85]	PR[84]	Res.	PR[82]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value										0	0	0		0																		
0xAC-0xBC	Reserved	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.